

Calc 3

10.6.20

11–20 ■ Find the traces of the given surface in the planes $x = k$, $y = k$, $z = k$. Then identify the surface and sketch it.

17. $x^2 + 4z^2 - y = 0$

18. $x^2 + 4y^2 + z^2 = 4$

19. $y = z^2 - x^2$

20. $16x^2 = y^2 + 4z^2$

11.6.40

- 40.** Find the points on the ellipsoid $x^2 + 2y^2 + 3z^2 = 1$ where the tangent plane is parallel to the plane $3x - y + 3z = 1$.

12.1.12

11–20 ■ Calculate the iterated integral.

11. $\int_1^3 \int_0^1 (1 + 4xy) \, dx \, dy$

12. $\int_2^4 \int_{-1}^1 (x^2 + y^2) \, dy \, dx$

